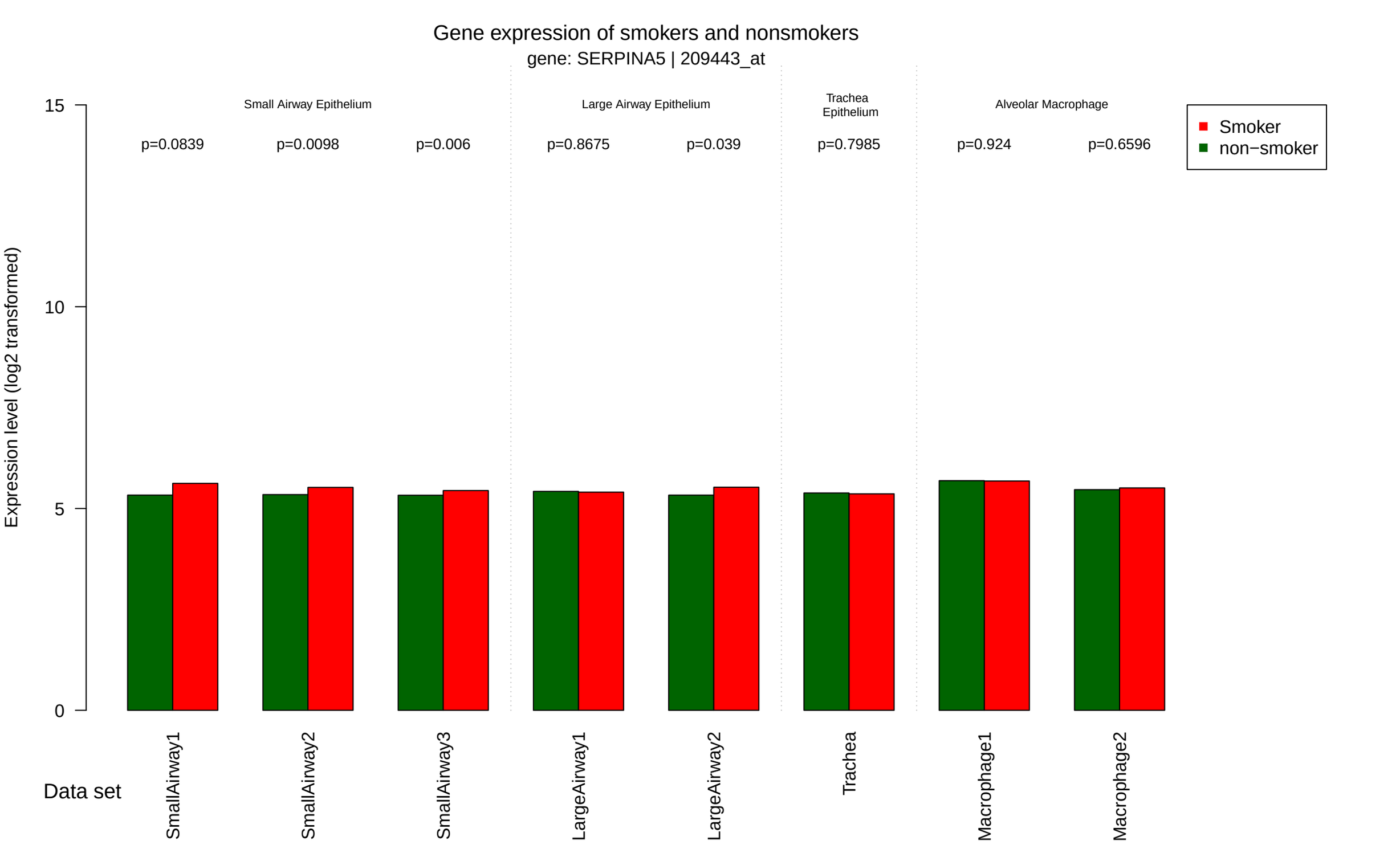
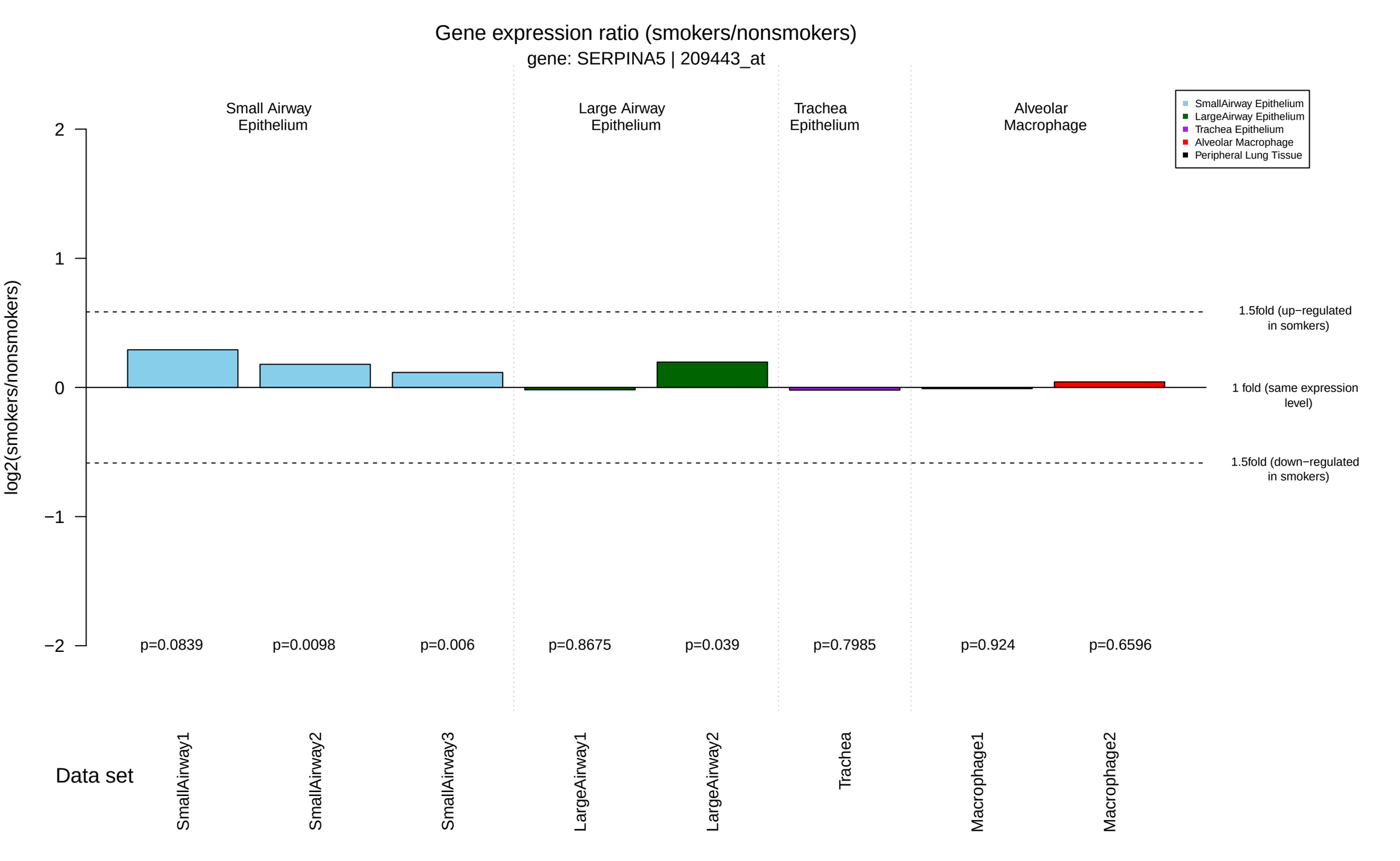
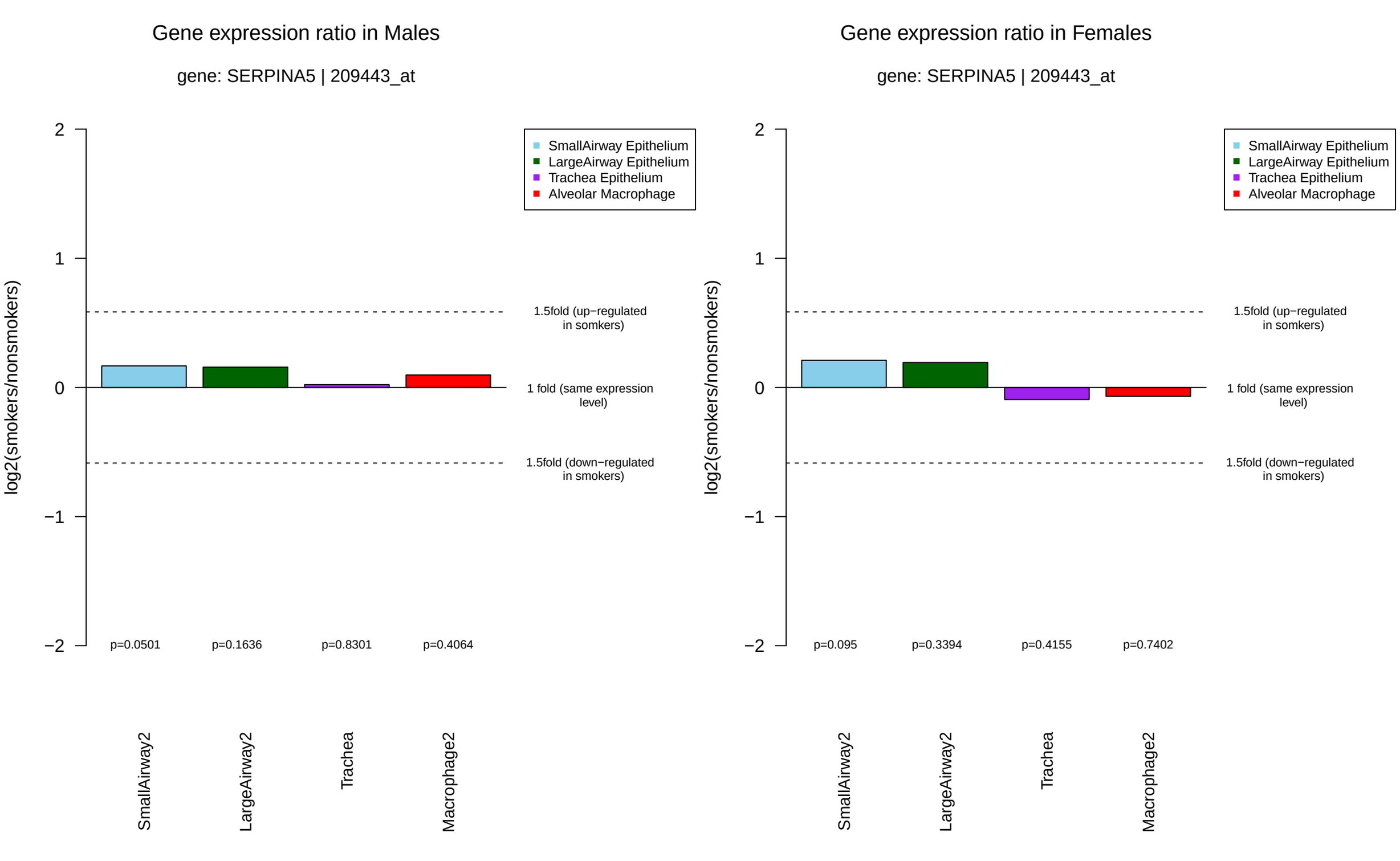
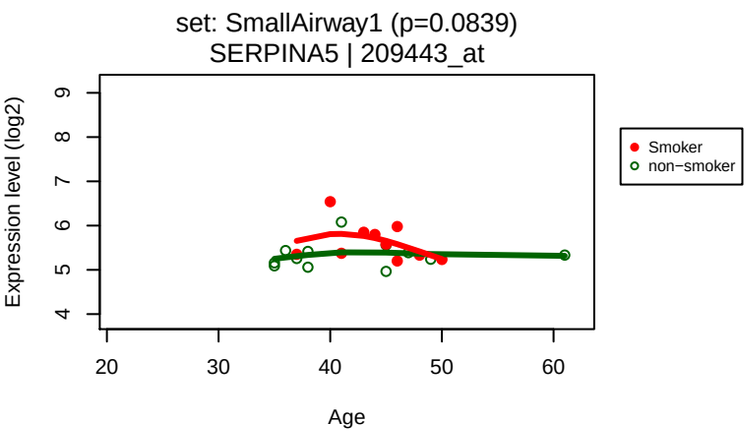
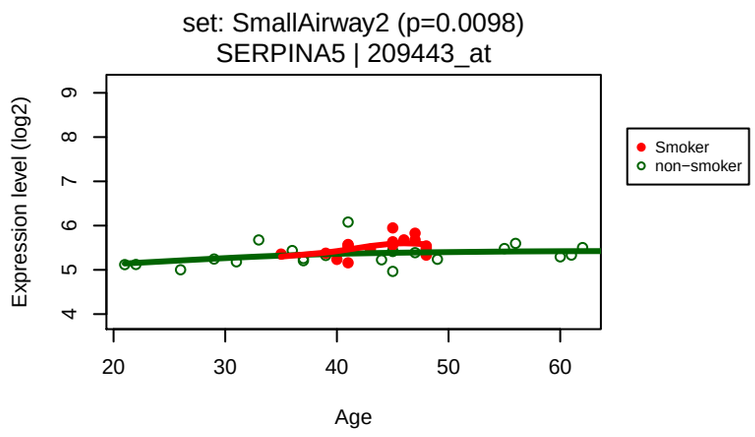
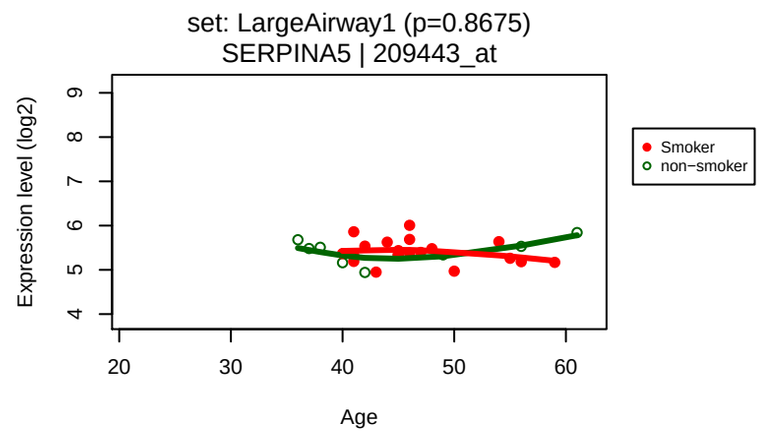

Gene expression of smokers and nonsmokers
gene: SERPINA5 | 209443_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0839
p=0.0098
p=0.006
p=0.8675
p=0.039
p=0.7985
p=0.924
p=0.6596
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

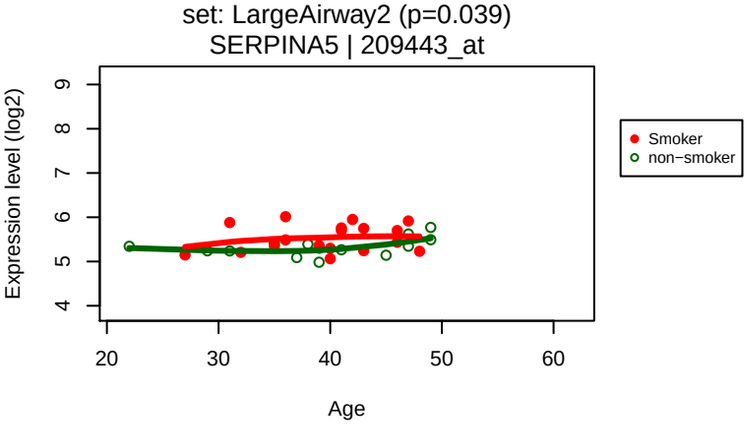

Gene expression ratio (smokers/nonsmokers)
gene: SERPINA5 | 209443_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0839
p=0.0098
p=0.006
p=0.8675
p=0.039
p=0.7985
p=0.924
p=0.6596
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

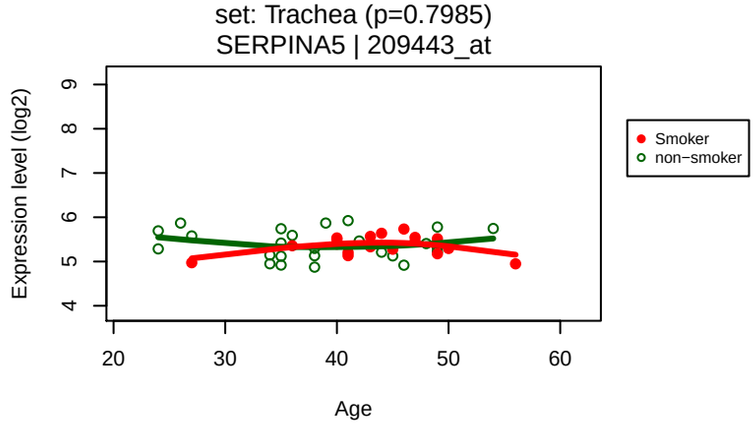

Gene expression ratio in Males
gene: SERPINA5 | 209443_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0501
p=0.1636
p=0.8301
p=0.4064
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINA5 | 209443_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.095
p=0.3394
p=0.4155
p=0.7402
SmallAirway2
LargeAirway2
Trachea
Macrophage2

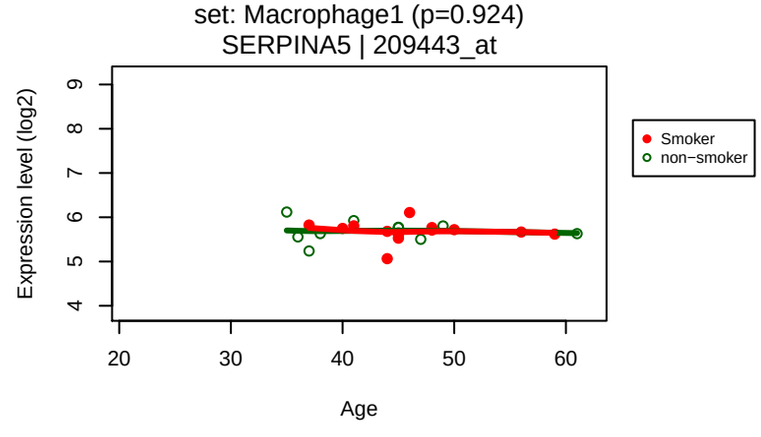

set: SmallAirway1 (p=0.0839)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker

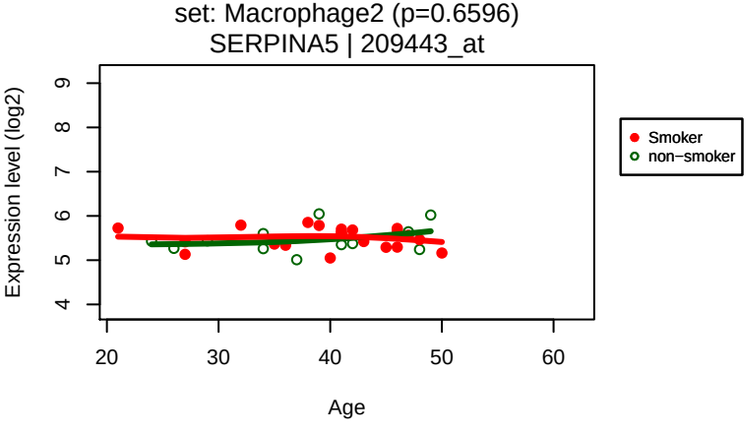

set: SmallAirway2 (p=0.0098)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker

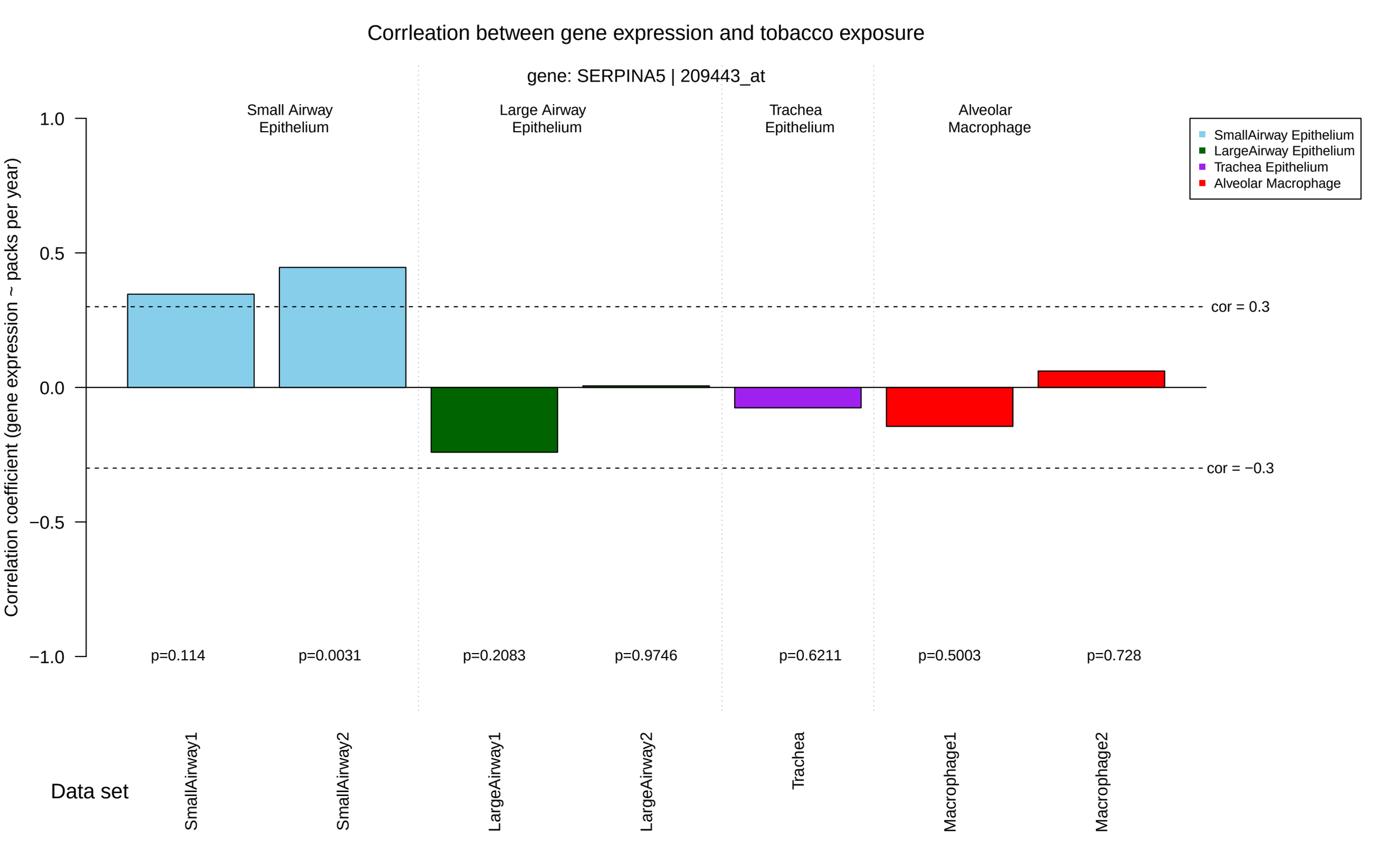

set: LargeAirway1 (p=0.8675)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.039)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7985)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.924)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6596)
SERPINA5 | 209443_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINA5 | 209443_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.114
p=0.0031
p=0.2083
p=0.9746
p=0.6211
p=0.5003
p=0.728
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2